Categorized Literature Review: AI■Driven Dinacharya Personalization

Category 1: AI and Machine Learning in Ayurveda

Salvi & Kadam (2025) – Prakriti Analysis Using AI: A Convergence of Ayurveda and Modern Technology
This paper surveys artificial intelligence approaches used for Prakriti classification in Ayurveda. It highlights how machine learning can reduce subjectivity in traditional Ayurvedic assessments that usually depend on practitioner expertise. The work is important for the Dinacharya automation project because accurate Prakriti identification is required to generate personalized daily routine recommendations.
Link: https://www.thevoiceofcreativeresearch.com/index.php/vcr/article/view/76

Trivedi & Patel (2025) – Human Prakriti Classification Using Machine Learning
This research proposes a machine learning pipeline using facial skin features to classify Ayurvedic constitution types. Algorithms such as Random Forest, KNN and SVM are used to analyze physiological traits automatically. The study supports the current project by demonstrating how computational models can automate constitution detection required for personalized Dinacharya planning.
Link: https://jisem-journal.com/index.php/journal/article/view/6973

Mallajosyula et al. (2025) – Artificial Intelligence and Machine Learning in Ayurveda
The paper explores how artificial intelligence can digitize and scale Ayurvedic knowledge systems. It discusses applications such as constitution analysis, herbal drug discovery and personalized medicine. This work supports the idea that AI systems can operationalize traditional healthcare frameworks such as Dinacharya using computational models.
Link: https://journals.lww.com/ijar/fulltext/2025/10000/artificial_intelligence_and_machine_learning_in.14.aspx

AI for Objective Prakriti Assessment (2025)
This work proposes a multimodal AI framework combining questionnaire data and physiological traits to determine Ayurvedic constitution. The system uses machine learning to create scalable diagnostic tools. This research directly supports the Dinacharya project because personalized daily routines depend on reliable automated Prakriti assessment.
Link: https://theaspd.com/index.php/ijes/article/download/5972/4322/12121

Category 2: Computational Models and Datasets for Prakriti

AI■Driven Dosha Analysis Survey (2025)
This paper explores machine learning systems designed to detect dosha imbalance and recommend preventive health strategies. The work emphasizes using AI to transform traditional medicine into data-driven healthcare systems. Such approaches support automated lifestyle recommendation engines based on Ayurvedic principles.
Link: https://ijarcce.com/wp-content/uploads/2025/06/IJARCCE.2025.14609.pdf

Comparative ML Algorithms for Ayurveda (2025)
This study compares multiple machine learning algorithms for identifying Ayurvedic constitution using health datasets. It demonstrates that physiological patterns can be

identified using computational models. The findings strengthen the feasibility of AI pipelines for constitution-aware health recommendation systems.
Link: https://www.sciencedirect.com/science/article/abs/pii/S3050475925003148

Prakriti200 Dataset for Ayurvedic Analysis (2025)
This research introduces a structured dataset containing physiological and psychological features mapped to Vata, Pitta and Kapha categories. Such datasets are essential for training machine learning models used in computational Ayurveda systems. The dataset helps enable automated health assistants capable of generating personalized recommendations.
Link: https://arxiv.org/abs/2510.06262

Psychometric ML-Based Prakriti Prediction (2025)
This research integrates behavioral questionnaire data with machine learning algorithms to determine Ayurvedic constitution. The study demonstrates that psychological and physiological traits together can improve classification accuracy. This supports the project's goal of combining multiple signals for personalized lifestyle recommendations.
Link: https://onlinelibrary.wiley.com/doi/10.1111/coin.70133

Category 3: AI■Driven Digital Health Systems

AI Digital Behavior Change Systems (2024)
This paper explores how artificial intelligence can guide behavior change using digital health platforms. It explains adaptive feedback mechanisms and personalized recommendations. These techniques are important for Dinacharya automation because the system must encourage users to follow structured daily routines consistently.
Link: https://www.mcpdigitalhealth.org/article/S2949-7612(24)00042-7/fulltext

Circadian Rhythm Digital Therapeutics (2025)
The study evaluates digital therapeutic systems that provide circadian-based recommendations using wearable sensors. It shows how daily behavior suggestions can adapt dynamically based on physiological data. This aligns closely with Dinacharya principles that organize activities according to biological time cycles.
Link: https://www.researchgate.net/publication/401548773

AI■Optimized Natural Product Systems (2025)
This research applies machine learning to analyze traditional medicine databases in order to identify therapeutic compounds. It demonstrates how AI can extract patterns from historical healthcare knowledge. Similar approaches can be used to digitize Ayurvedic lifestyle and wellness recommendations.
Link: https://arxiv.org/abs/2505.09643

Medicinal Plant Recognition using Deep Learning (2025)
The study proposes a convolutional neural network for identifying medicinal plants used in traditional medicine. Image recognition models classify herbs with high accuracy. This research illustrates how AI can digitize and preserve traditional healthcare knowledge systems.
Link: https://arxiv.org/abs/2501.09363

Category 4: Circadian Rhythm and Lifestyle Science

Xin et al. (2025) – Circadian Rhythm as a Target for Health Interventions
This study explains how circadian rhythm alignment improves metabolic and neurological health outcomes. It highlights the role of biological clocks in regulating daily physiological processes. These findings provide a scientific explanation for time-based lifestyle frameworks such as the Ayurvedic Dinacharya system.
Link: https://www.sciencedirect.com/science/article/pii/S2090123224001334

Sutton et al. (2025) – Exercise Timing and Circadian Regulation
The research investigates how exercise timing influences metabolic efficiency and physiological performance. It shows that aligning physical activity with biological rhythms improves health outcomes. These findings correspond with Ayurvedic recommendations that specify appropriate times for exercise.
Link: https://www.mdpi.com/2076-3921/14/9/1132

Robinson et al. (2025) – Circadian Lifestyle Interventions
This research studies lifestyle interventions designed to stabilize circadian rhythms. It demonstrates that structured daily routines improve metabolic and psychological health. The findings parallel Ayurvedic daily regimen systems such as Dinacharya.
Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC12608408/

Circadian Clock and Aging Study (2025)
The study explores how circadian rhythm disruptions affect metabolic regulation and aging. Biological clocks were shown to influence multiple organ systems. The work strengthens the scientific foundation for routine-based health frameworks that align activities with time.
Link: https://www.frontiersin.org/articles/10.3389/fphys.2025.1654369

Category 5: Supporting Biological Foundations

Circadian Rhythm Disruption and Mental Health (2020)
This research examines how disruption of biological clocks affects neurological and psychological health outcomes. It highlights the importance of consistent daily sleep and activity cycles. The findings support structured lifestyle systems such as Dinacharya.
Link: https://en.wikipedia.org/wiki/Chronodisruption

Circadian Rhythm and Human Physiology Review (2020)
This work explains how circadian rhythms regulate sleep, hormone release and metabolism. Maintaining stable daily behavior patterns was shown to improve physiological stability. These findings provide a modern biological explanation for Ayurvedic daily routine practices.
Link: https://en.wikipedia.org/wiki/Circadian_rhythm

CNN■Based Herbal Identification App (2025)
This research develops a mobile application that identifies medicinal herbs using deep learning models. The system demonstrates how AI can support traditional medicine practice. Such digital tools complement broader AI-driven Ayurveda healthcare platforms.
Link: https://arxiv.org/abs/2505.02147

Chronobiology and Lifestyle Regulation Study (2023)

This study investigates how daily lifestyle timing influences metabolic and cognitive health. The results show that maintaining regular activity cycles improves overall physiological function. These findings support the core principle of Dinacharya which structures daily activities around biological rhythms.
Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC6123576/